Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Frank Gwerder Dairy, Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
1007 S Hart RD	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 02/01/1964

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X012-X050-X005-XXXX X012-X051-X011-XXXX

B. OPERATORS

Gwerder, Frank

Operator name: Gwerder, Frank

Telephone no.: (209) 579-5723 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
825 S Hart RD	Modesto	CA	95358

This operator is responsible for paying permit fees.

C. OWNERS

Glattke, Marlene

Legal owner name: Glattke, Marlene

Telephone no.: (209) 579-5723 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
825 S Hart RD	Modesto	CA	95358

Gwerder, Frank

Legal owner name: Gwerder, Frank

Telephone no.: (209) 579-5723

Landline	Cellular
(209) 579-5723	

Mailing Address Number and Street	City	State	Zip Code
825 S Hart RD	Modesto	CA	95358

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	250	100	0
Number under roof	760	100	175	0	0	0
Maximum number	760	100	175	250	100	0
Average number	760	100	175	250	100	0
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 25,858.28 *tons per reporting period*

Total nitrogen from manure: 323,246.13 *lbs per reporting period*

After ammonia losses (30% loss applied): 226,272.29 *lbs per reporting period*

Total phosphorus from manure: 53,604.72 *lbs per reporting period*

Total potassium from manure: 147,544.29 *lbs per reporting period*

Total salt from manure: 380,841.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,893,788 *gallons*

Total nitrogen generated: 113,407.90 *lbs*

Total phosphorus generated: 7,873.84 *lbs*

Total potassium generated: 112,540.65 *lbs*

Total salt generated: 647,139.84 *lbs*

	8,560,017	*gallons applied*
+	4,333,771	*gallons exported*
-	0	*gallons imported*
=	12,893,788	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 4	Surface water
TID Turlock Main	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
08/18/2023	Solid commercial fertilizer *UN32*	15.76 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	10,076.31	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	10,076.31	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/30/2023	Corral solids	1,640.00 *ton*	Dry-weight	19.6		18,500.00	10,200.00	48,000.00		45.50

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
04/09/2023	Process wastewater	812,596.00 *gal*	1,540.00	546.00	0.00	0.00	82.40	1,160.00		5,100
05/25/2023	Process wastewater	956,462.00 *gal*	1,540.00	546.00	0.00	0.00	82.40	1,160.00		5,100
07/22/2023	Process wastewater	934,587.00 *gal*	501.00	333.00	0.00	0.00	36.70	1,160.00		9,280
08/04/2023	Process wastewater	917,626.00 *gal*	501.00	333.00	0.00	0.00	36.70	1,160.00		9,280
10/23/2023	Process wastewater sludge	712,500.00 *gal*	818.00	126.00	0.00	0.00	66.40	700.00		1,660

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	48,786.72	26,898.62	126,581.76	1,199,889.60
Process wastewater	35,342.19	2,178.52	39,216.70	228,598.61
Total exports for all materials	84,128.91	29,077.14	165,798.46	1,428,488.21

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
A2	23	23	2	manure	X012-X050-X008-XXXX
B1	20	20	2	process wastewater	X012-X051-X011-XXXX
B2	13	13	2	process wastewater	X012-X051-X011-XXXX
B3	9	9	2	process wastewater	X012-X051-X010-XXXX
B4	11	11	2	process wastewater	X012-X051-X011-XXXX
B5	4	4	2	process wastewater	X012-X051-X011-XXXX
B6	1	1	2	process wastewater	X012-X051-X011-XXXX
B7	75	75	2	process wastewater	X012-X051-X006-XXXX
B8	38	38	1	manure	X017-X002-X002-XXXX
C1	180	89	2	manure	X080-X014-X008-XXXX
Totals for areas that were used for application	374	283	19		
Totals for areas that were not used for application					
Land application area totals	374	283	19		

B. CROPS AND HARVESTS

A2

Field name: A2

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	413.21 *ton*	As-is		72.6	3,650.00	830.00	4,940.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.97	131.15	29.82	177.50	1,151.89

A2

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	589.00 *ton*	As-is		62.6	4,260.00	830.00	4,000.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	120.00	22.50	99.00	0.00
Total actual harvest content	25.61	218.19	42.51	204.87	976.92

B1

Field name: B1

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	343.00 *ton*	As-is		73.2	5,640.00	850.00	6,750.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.15	193.45	29.16	231.53	1,075.51

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	506.00 *ton*	As-is		60.6	4,930.00	800.00	6,130.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.30	249.46	40.48	310.18	996.82

B2

Field name: B2

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	214.00 *ton*	As-is		73.2	5,640.00	8,500.00	67,500.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.46	185.69	279.85	2,222.31	1,032.34

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	355.00 *ton*	As-is		60.6	4,930.00	800.00	6,130.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.31	269.25	43.69	334.79	1,075.92

B3

Field name: B3

11/04/2002: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 11/04/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	146.00 *ton*	As-is		73.2	5,640.00	850.00	6,750.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.22	182.99	27.58	219.00	1,017.33

B3

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	248.00 *ton*	As-is		60.6	4,930.00	800.00	6,130.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.56	271.70	44.09	337.83	1,085.69

B4

Field name: B4

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	175.00 *ton*	As-is		73.2	5,640.00	850.00	6,750.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.91	179.45	27.05	214.77	997.69

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	269.00 *ton*	As-is		60.6	4,930.00	800.00	6,130.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.45	241.12	39.13	299.81	963.51

B5

Field name: B5

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 4 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	63.00 *ton*	As-is		73.2	5,640.00	850.00	6,750.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	22.50	99.00	0.00
Total actual harvest content	15.75	177.66	26.78	212.63	987.71

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 4 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	101.00 *ton*	As-is		60.6	4,930.00	800.00	6,130.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.25	248.97	40.40	309.57	994.85

B6

Field name: B6

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 1 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	16.16 *ton*	As-is		73.2	5,640.00	850.00	6,750.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.16	182.28	27.47	218.16	1,013.43

B6

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 1 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	26.00 *ton*	As-is		60.6	4,930.00	800.00	6,130.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	256.36	41.60	318.76	1,024.40

B7

Field name: B7

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	1,455.04 *ton*	As-is		71.6	5,340.00	840.00	5,810.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	19.40	207.20	32.59	225.43	1,079.91

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,988.00 *ton*	As-is		63.2	4,670.00	960.00	5,110.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.51	247.57	50.89	270.90	1,346.11

B8

Field name: B8

02/12/2019: Almond, in shell

Crop: Almond, in shell Acres planted: 38 Plant date: 02/12/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	153.00 *ton*	As-is		6.2	25,500.00	1,810.00	25,700.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	215.00	36.50	232.50	0.00
Total actual harvest content	4.03	205.34	14.58	206.95	785.55

C1

Field name: C1

11/18/2022: Oats, hay

Crop: Oats, hay Acres planted: 89 Plant date: 11/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	566.00 *ton*	As-is		9.6	8,050.00	1,110.00	11,300.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	6.36	102.39	14.12	143.73	1,080.82

06/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 89 Plant date: 06/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	2,382.00 *ton*	As-is		63.4	4,170.00	880.00	4,500.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.76	223.21	47.10	240.88	1,097.11

NUTRIENT BUDGET

A. LAND APPLICATIONS

A2 - 10/27/2022: Oats, silage-soft dough

Field name: A2

Crop: Oats, silage-soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	0.00	3,825,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator SOlids	Separator solids	169.90	127.22	244.39	3,783.04	350.00 *ton*
Application event totals		169.90	127.22	244.39	3,783.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	49.22	3,570,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.22	

A2 - 06/15/2023: Corn, silage

Field name: A2

Crop: Corn, silage — Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Soldis	Corral solids	226.10	69.19	231.20	12,172.00	230.00 *ton*
Application event totals		226.10	69.19	231.20	12,172.00	

A2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	56.25	4,080,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	49.22	3,570,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	45.71	3,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	49.22	3,570,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.22	

A2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	49.22	3,570,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	49.22	3,570,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	49.22	3,570,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.22	

B1 - 11/04/2022: Oats, silage-soft dough

Field name: B1

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	228.32	15.58	76.11	563.66	285,000.00 *gal*
Application event totals		228.32	15.58	76.11	563.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	52.56	3,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	52.56	

B1 - 06/01/2023: Corn, silage

Field name: B1

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	263.21	14.08	198.26	871.67	409,626.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	52.56	3,315,000.00 *gal*
Application event totals		263.21	14.08	198.26	924.23	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	42.45	2,677,500.00 *gal*
Application event totals		0.00	0.00	0.00	42.45	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	2,550,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.61	5.61	177.37	1,418.98	366,465.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	2,550,000.00 *gal*
Application event totals		76.61	5.61	177.37	1,459.41	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	2,550,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

B1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	2,550,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

B2 - 11/04/2022: Oats, silage-soft dough

Field name: B2

Crop: Oats, silage-soft dough Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	228.32	15.58	76.11	563.66	185,250.00 *gal*
Application event totals		228.32	15.58	76.11	563.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	49.76	2,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.76	

B2 - 06/01/2023: Corn, silage

Field name: B2

Crop: Corn, silage Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

B2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	259.15	13.87	195.20	858.23	262,149.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	52.87	2,167,500.00 *gal*
Application event totals		259.15	13.87	195.20	911.10	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	46.65	1,912,500.00 *gal*
Application event totals		0.00	0.00	0.00	46.65	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	43.54	1,785,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.54	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.05	6.01	189.97	1,519.74	255,116.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	37.32	1,530,000.00 *gal*
Application event totals		82.05	6.01	189.97	1,557.06	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	43.54	1,785,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.54	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	37.32	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.32	

B3 - 11/04/2002: Oats, silage-soft dough

Field name: B3

Crop: Oats, silage-soft dough

Plant date: 11/04/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	228.32	15.58	76.11	563.66	128,250.00 *gal*
Application event totals		228.32	15.58	76.11	563.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	53.91	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.91	

B3 - 06/01/2023: Corn, silage

Field name: B3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	259.25	13.87	195.28	858.55	181,556.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	58.40	1,657,500.00 *gal*
Application event totals		259.25	13.87	195.28	916.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	53.91	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.91	

B3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	53.91	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.90	5.78	182.68	1,461.46	169,846.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	53.91	1,530,000.00 *gal*
Application event totals		78.90	5.78	182.68	1,515.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	53.91	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	44.92	1,275,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.92	

B4 - 11/04/2022: Oats, silage-soft dough

Field name: B4

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	228.32	15.58	76.11	563.66	156,750.00 *gal*
Application event totals		228.32	15.58	76.11	563.66	

B4 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	51.46	1,785,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.46	

B4 - 06/01/2023: Corn, silage

Field name: B4

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	256.44	13.72	193.16	849.24	219,495.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	58.81	2,040,000.00 *gal*
Application event totals		256.44	13.72	193.16	908.05	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	47.78	1,657,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.78	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	47.78	1,657,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.78	

B4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.10	5.72	180.83	1,446.67	205,489.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	44.11	1,530,000.00 *gal*
Application event totals		78.10	5.72	180.83	1,490.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	47.78	1,657,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	44.11	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.11	

B5 - 11/04/2022: Oats, silage-soft dough

Field name: B5

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	228.32	15.58	76.11	563.66	57,000.00 *gal*
Application event totals		228.32	15.58	76.11	563.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	50.54	637,500.00 *gal*
Application event totals		0.00	0.00	0.00	50.54	

B5 - 06/01/2023: Corn, silage

Field name: B5

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	257.55	13.78	194.00	852.91	80,162.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	60.65	765,000.00 *gal*
Application event totals		257.55	13.78	194.00	913.56	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	50.54	637,500.00 *gal*
Application event totals		0.00	0.00	0.00	50.54	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	50.54	637,500.00 *gal*
Application event totals		0.00	0.00	0.00	50.54	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.01	5.86	185.25	1,481.96	76,546.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	510,000.00 *gal*
Application event totals		80.01	5.86	185.25	1,522.39	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	510,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

B5 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	510,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

B6 - 11/04/2022: Oats, silage-soft dough

Field name: B6

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	228.32	15.58	76.11	563.66	14,250.00 *gal*
Application event totals		228.32	15.58	76.11	563.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	80.86	255,000.00 *gal*
Application event totals		0.00	0.00	0.00	80.86	

B6 - 06/01/2023: Corn, silage

Field name: B6

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

B6 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	262.76	14.06	197.92	870.17	20,446.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	80.86	255,000.00 *gal*
Application event totals		262.76	14.06	197.92	951.03	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	80.86	255,000.00 *gal*
Application event totals		0.00	0.00	0.00	80.86	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	127,500.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.97	6.08	192.11	1,536.91	19,846.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	127,500.00 *gal*
Application event totals		82.97	6.08	192.11	1,577.34	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	127,500.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	40.43	127,500.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

B7 - 10/26/2022: Oats, silage-soft dough

Field name: B7

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.35	15.71	97.75	378.21	1,249,669.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	51.75	12,240,000.00 *gal*
Application event totals		65.35	15.71	97.75	429.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.04	6.64	67.59	358.07	646,216.00 *gal*
Application event totals		71.04	6.64	67.59	358.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	144.91	7.75	109.15	479.88	845,666.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	47.44	11,220,000.00 *gal*
Application event totals		144.91	7.75	109.15	527.32	

B7 - 05/31/2023: Corn, silage

Field name: B7

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	250.61	13.41	188.77	829.95	1,462,565.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	52.83	12,495,000.00 *gal*
Application event totals		250.61	13.41	188.77	882.78	

B7 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	49.60	11,730,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.39	5.16	162.97	1,303.76	1,262,659.00 *gal*
MID Lateral 4	Surface water	0.00	0.00	0.00	49.60	11,730,000.00 *gal*
Application event totals		70.39	5.16	162.97	1,353.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	47.44	11,220,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	47.44	11,220,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	43.13	10,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.13	

B8 - 02/12/2019: Almond, in shell

Field name: B8

Crop: Almond, in shell

Plant date: 02/12/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

B8 - 02/12/2019: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	0.00	3,562,699.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

11/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	70.52	52.80	101.43	1,570.11	240.00 *ton*
Application event totals		70.52	52.80	101.43	1,570.11	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	0.00	3,562,699.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

04/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	59.46	7,125,398.00 *gal*
Application event totals		0.00	0.00	0.00	59.46	

05/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

B8 - 02/12/2019: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	59.46	7,125,398.00 *gal*
Application event totals		0.00	0.00	0.00	59.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	59.46	7,125,398.00 *gal*
Application event totals		0.00	0.00	0.00	59.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	59.46	7,125,398.00 *gal*
Application event totals		0.00	0.00	0.00	59.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 4	Surface water	0.00	0.00	0.00	29.73	3,562,699.00 *gal*
Application event totals		0.00	0.00	0.00	29.73	

C1 - 11/18/2022: Oats, hay

Field name: C1

Crop: Oats, hay

Plant date: 11/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator SOlids	Separator solids	139.25	104.27	200.30	3,100.52	1,110.00 *ton*
Application event totals		139.25	104.27	200.30	3,100.52	

C1 - 11/18/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	0.00	14,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	16.16	13,260,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.16	

C1 - 06/21/2023: Corn, silage

Field name: C1

Crop: Corn, silage

Plant date: 06/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	231.18	70.74	236.40	12,445.53	910.00 *ton*
Application event totals		231.18	70.74	236.40	12,445.53	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	20.51	16,830,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.51	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	18.65	15,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.65	

C1 - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	18.65	15,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.65	

08/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	18.65	15,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.65	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	17.41	14,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.41	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	18.03	14,790,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.03	

C1 - 06/21/2023: Corn, silage

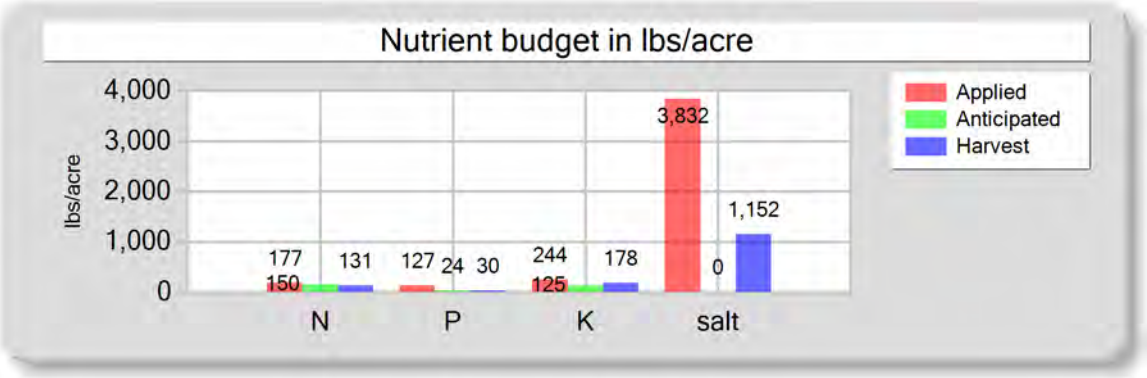
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Turlock Main	Surface water	0.00	0.00	0.00	17.41	14,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.41	

B. NUTRIENT BUDGET

A2 - 10/27/2022: Oats, silage-soft dough

Field name: A2 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.90	127.22	244.39	3,783.04
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	49.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.90	127.22	244.39	3,832.26
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	131.15	29.82	177.50	1,151.89
Nutrient balance	45.75	97.39	66.89	2,680.38
Applied to removed ratio	1.35	4.27	1.38	3.33

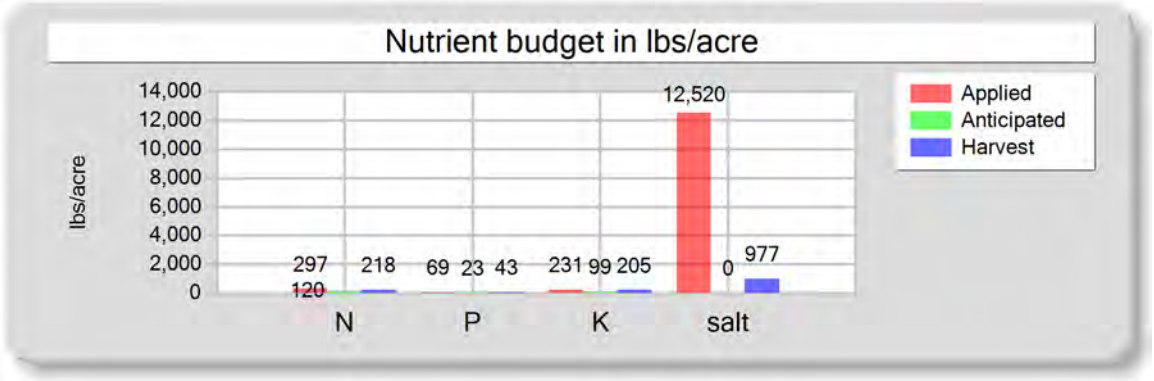
Fresh water applied	
7,395,000.00	*gallons*
272.33	*acre-inches*
11.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

A2 - 06/15/2023: Corn, silage

Field name: A2 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	226.10	69.19	231.20	12,172.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	348.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.10	69.19	231.20	12,520.06
Anticipated crop nutrient removal	120.00	22.50	99.00	0.00
Actual crop nutrient removal	218.19	42.51	204.87	976.92
Nutrient balance	78.91	26.68	26.33	11,543.14
Applied to removed ratio	1.36	1.63	1.13	12.82

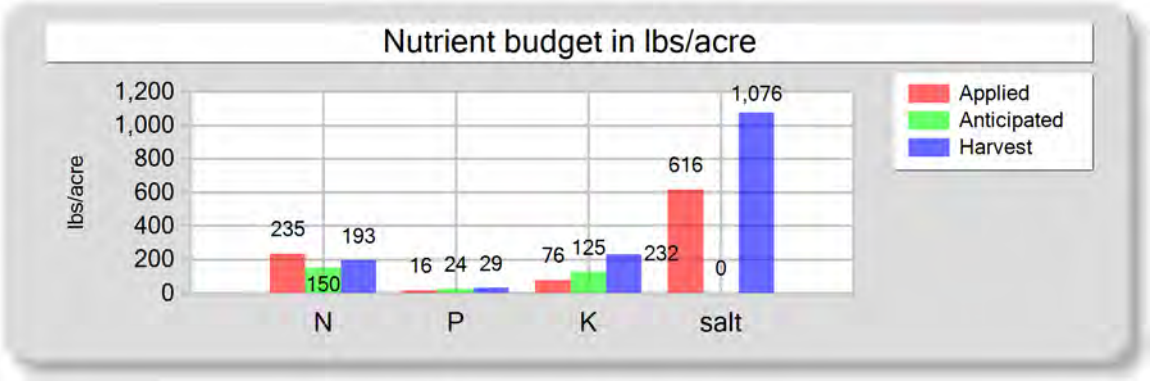
Fresh water applied	
25,245,000.00	*gallons*
929.69	*acre-inches*
40.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B1 - 11/04/2022: Oats, silage-soft dough

Field name: B1 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	15.58	76.11	563.66
Fresh water	0.00	0.00	0.00	52.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.32	15.58	76.11	616.22
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	193.45	29.16	231.53	1,075.51
Nutrient balance	41.87	-13.58	-155.42	-459.29
Applied to removed ratio	1.22	0.53	0.33	0.57

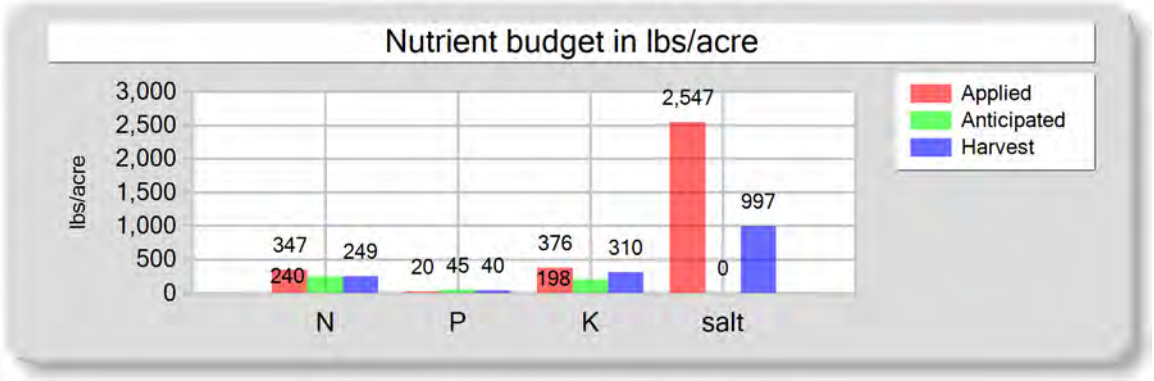
Fresh water applied	
3,315,000.00	*gallons*
122.08	*acre-inches*
6.10	*inches/acre*

Process wastewater applied	
285,000.00	*gallons*
10.50	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

B1 - 06/01/2023: Corn, silage

Field name: B1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	339.82	19.70	375.64	2,290.66
Fresh water	0.00	0.00	0.00	256.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	346.82	19.70	375.64	2,547.40
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.46	40.48	310.18	996.82
Nutrient balance	97.36	-20.78	65.46	1,550.58
Applied to removed ratio	1.39	0.49	1.21	2.56

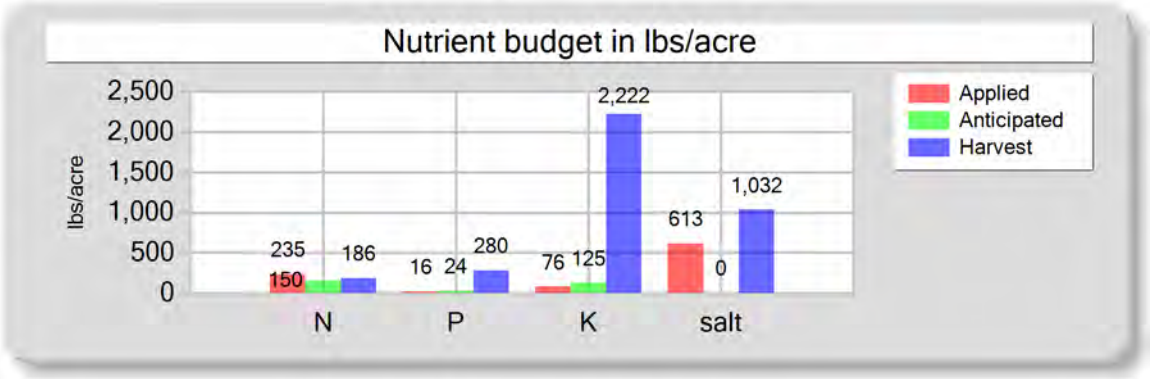
Fresh water applied	
16,192,500.00	*gallons*
596.31	*acre-inches*
29.82	*inches/acre*

Process wastewater applied	
776,091.00	*gallons*
28.58	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

B2 - 11/04/2022: Oats, silage-soft dough

Field name: B2 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	15.58	76.11	563.66
Fresh water	0.00	0.00	0.00	49.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.32	15.58	76.11	613.42
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	185.69	279.85	2,222.31	1,032.34
Nutrient balance	49.63	-264.27	-2,146.20	-418.91
Applied to removed ratio	1.27	0.06	0.03	0.59

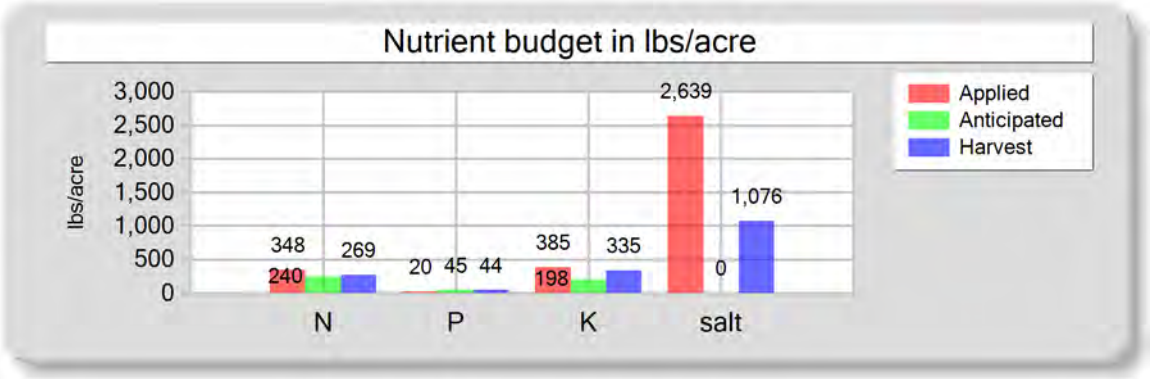
Fresh water applied	
2,040,000.00	*gallons*
75.13	*acre-inches*
5.78	*inches/acre*

Process wastewater applied	
185,250.00	*gallons*
6.82	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

B2 - 06/01/2023: Corn, silage

Field name: B2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	341.20	19.88	385.17	2,377.96
Fresh water	0.00	0.00	0.00	261.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	348.20	19.88	385.17	2,639.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	269.25	43.69	334.79	1,075.92
Nutrient balance	78.94	-23.82	50.38	1,563.29
Applied to removed ratio	1.29	0.45	1.15	2.45

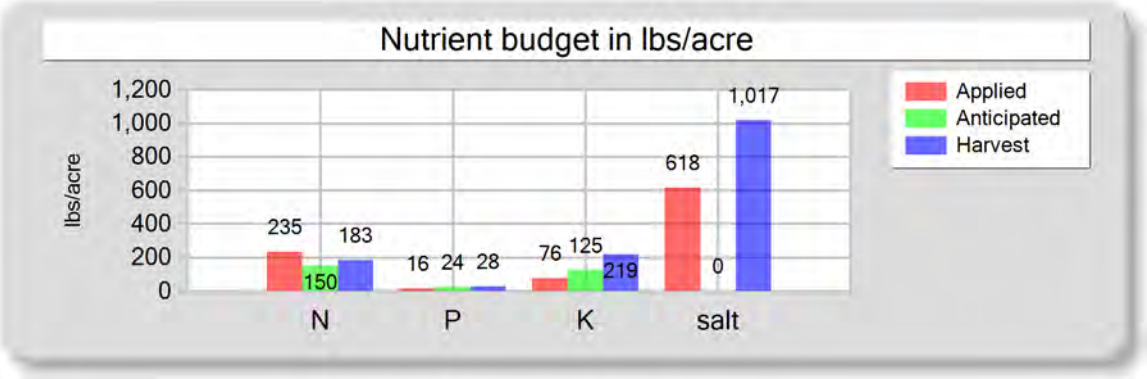
Fresh water applied	
10,710,000.00	*gallons*
394.41	*acre-inches*
30.34	*inches/acre*

Process wastewater applied	
517,265.00	*gallons*
19.05	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

B3 - 11/04/2002: Oats, silage-soft dough

Field name: B3 Crop: Oats, silage-soft dough Plant date: 11/04/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	15.58	76.11	563.66
Fresh water	0.00	0.00	0.00	53.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.32	15.58	76.11	617.57
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	182.99	27.58	219.00	1,017.33
Nutrient balance	52.33	-12.00	-142.89	-399.76
Applied to removed ratio	1.29	0.56	0.35	0.61

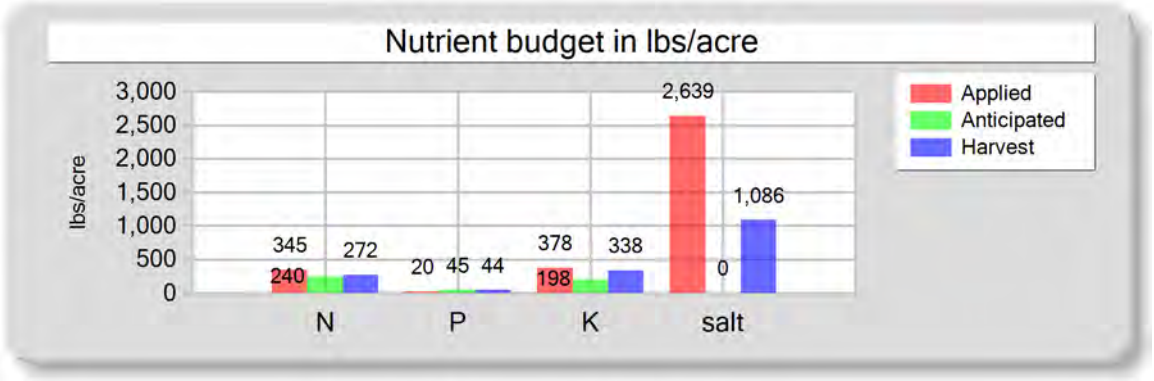
Fresh water applied	
1,530,000.00	*gallons*
56.34	*acre-inches*
6.26	*inches/acre*

Process wastewater applied	
128,250.00	*gallons*
4.72	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

B3 - 06/01/2023: Corn, silage

Field name: B3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	338.15	19.65	377.96	2,320.01
Fresh water	0.00	0.00	0.00	318.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	345.15	19.65	377.96	2,638.97
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	271.70	44.09	337.83	1,085.69
Nutrient balance	73.45	-24.44	40.13	1,553.28
Applied to removed ratio	1.27	0.45	1.12	2.43

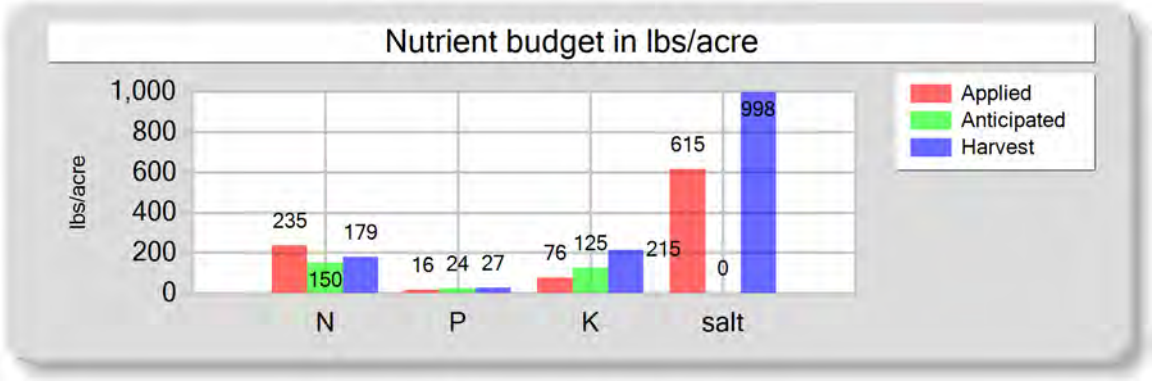
Fresh water applied	
9,052,500.00	*gallons*
333.37	*acre-inches*
37.04	*inches/acre*

Process wastewater applied	
351,402.00	*gallons*
12.94	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

B4 - 11/04/2022: Oats, silage-soft dough

Field name: B4 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	15.58	76.11	563.66
Fresh water	0.00	0.00	0.00	51.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.32	15.58	76.11	615.12
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	179.45	27.05	214.77	997.69
Nutrient balance	55.86	-11.47	-138.67	-382.57
Applied to removed ratio	1.31	0.58	0.35	0.62

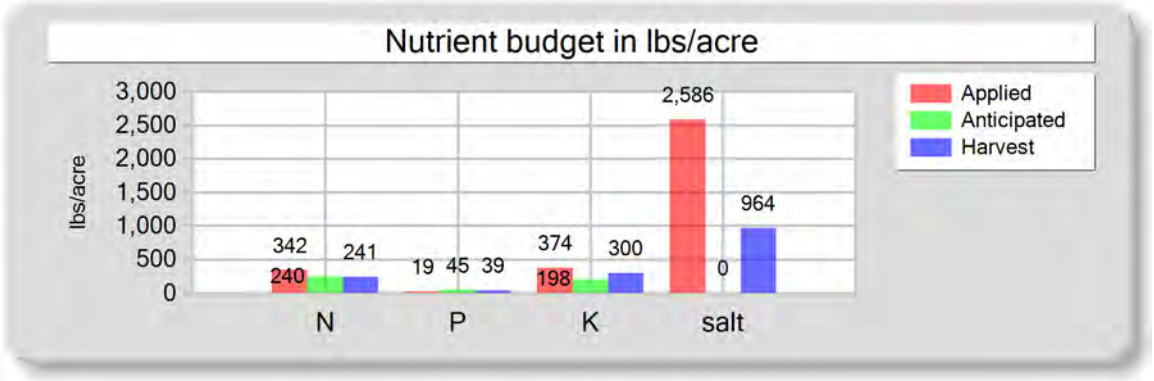
Fresh water applied	
1,785,000.00	*gallons*
65.74	*acre-inches*
5.98	*inches/acre*

Process wastewater applied	
156,750.00	*gallons*
5.77	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

B4 - 06/01/2023: Corn, silage

Field name: B4 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	334.54	19.44	373.99	2,295.91
Fresh water	0.00	0.00	0.00	290.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	341.54	19.44	373.99	2,586.28
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	241.12	39.13	299.81	963.51
Nutrient balance	100.42	-19.69	74.18	1,622.77
Applied to removed ratio	1.42	0.50	1.25	2.68

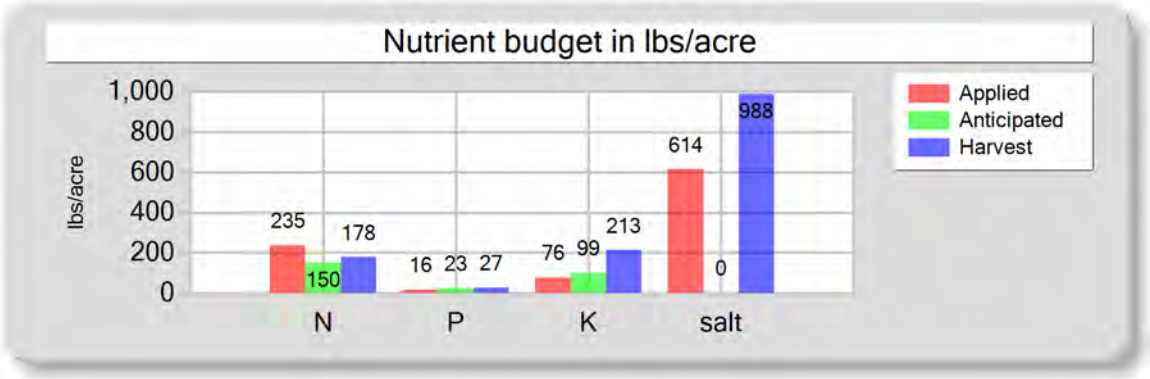
Fresh water applied	
10,072,500.00	*gallons*
370.94	*acre-inches*
33.72	*inches/acre*

Process wastewater applied	
424,984.00	*gallons*
15.65	*acre-inches*
1.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

B5 - 11/04/2022: Oats, silage-soft dough

Field name: B5 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	15.58	76.11	563.66
Fresh water	0.00	0.00	0.00	50.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.32	15.58	76.11	614.20
Anticipated crop nutrient removal	150.00	22.50	99.00	0.00
Actual crop nutrient removal	177.66	26.78	212.63	987.71
Nutrient balance	57.66	-11.20	-136.52	-373.51
Applied to removed ratio	1.32	0.58	0.36	0.62

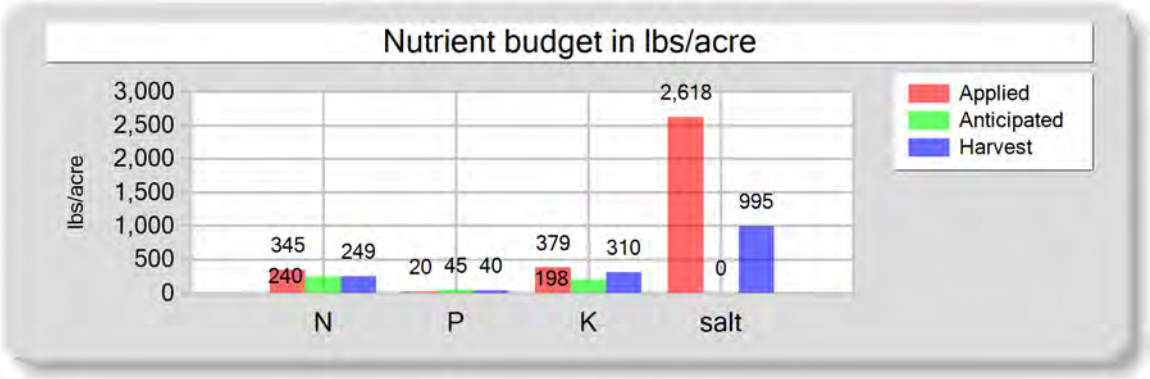
Fresh water applied	
637,500.00	*gallons*
23.48	*acre-inches*
5.87	*inches/acre*

Process wastewater applied	
57,000.00	*gallons*
2.10	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

B5 - 06/01/2023: Corn, silage

Field name: B5 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	337.55	19.64	379.24	2,334.87
Fresh water	0.00	0.00	0.00	283.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.55	19.64	379.24	2,617.90
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	248.97	40.40	309.57	994.85
Nutrient balance	95.59	-20.76	69.68	1,623.05
Applied to removed ratio	1.38	0.49	1.23	2.63

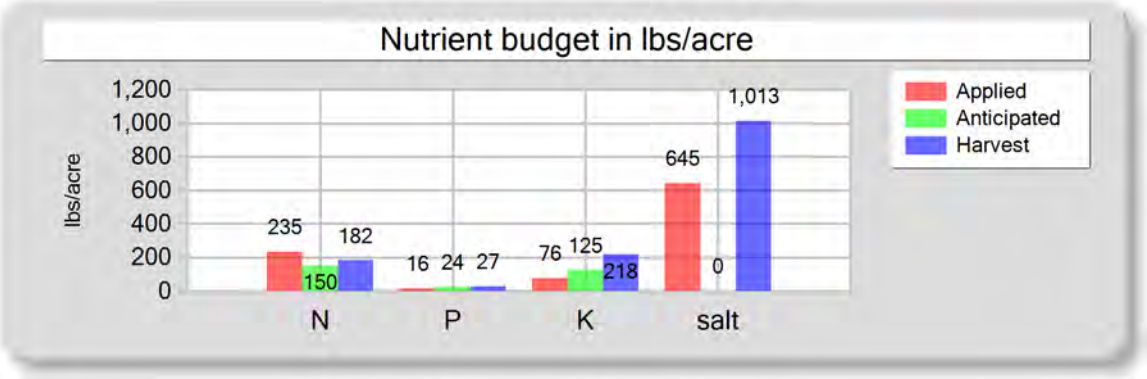
Fresh water applied	
3,570,000.00	*gallons*
131.47	*acre-inches*
32.87	*inches/acre*

Process wastewater applied	
156,708.00	*gallons*
5.77	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

B6 - 11/04/2022: Oats, silage-soft dough

Field name: B6 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.32	15.58	76.11	563.66
Fresh water	0.00	0.00	0.00	80.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.32	15.58	76.11	644.53
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	182.28	27.47	218.16	1,013.43
Nutrient balance	53.03	-11.89	-142.05	-368.90
Applied to removed ratio	1.29	0.57	0.35	0.64

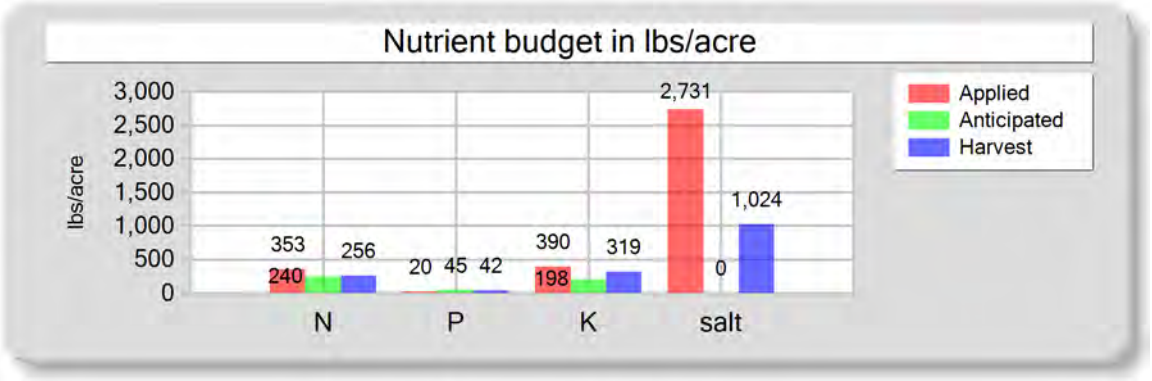
Fresh water applied
255,000.00 *gallons*
9.39 *acre-inches*
9.39 *inches/acre*

Process wastewater applied
14,250.00 *gallons*
0.52 *acre-inches*
0.52 *inches/acre*

Total harvests for the crop
1 *harvests*

B6 - 06/01/2023: Corn, silage

Field name: B6 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	345.73	20.14	390.03	2,407.08
Fresh water	0.00	0.00	0.00	323.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	352.73	20.14	390.03	2,730.53
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	256.36	41.60	318.76	1,024.40
Nutrient balance	96.37	-21.46	71.27	1,706.13
Applied to removed ratio	1.38	0.48	1.22	2.67

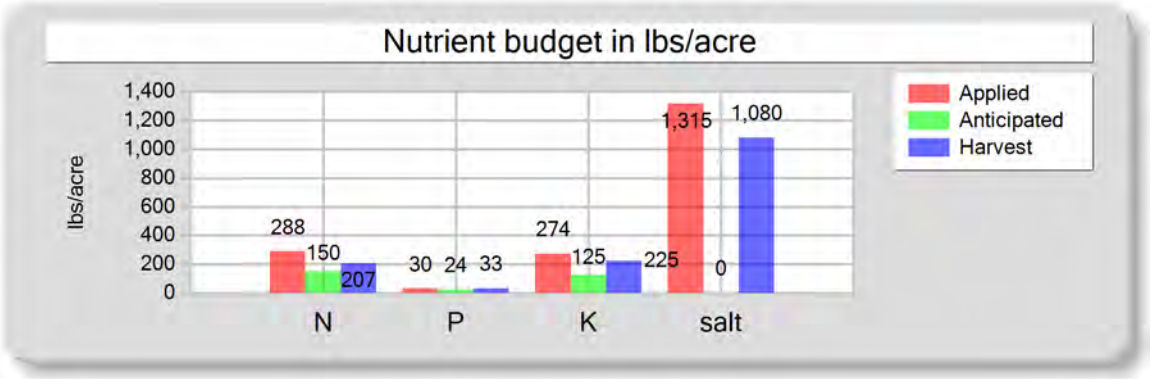
Fresh water applied	
1,020,000.00	*gallons*
37.56	*acre-inches*
37.56	*inches/acre*

Process wastewater applied	
40,292.00	*gallons*
1.48	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

B7 - 10/26/2022: Oats, silage-soft dough

Field name: B7 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	281.30	30.10	274.49	1,216.16
Fresh water	0.00	0.00	0.00	99.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.30	30.10	274.49	1,315.35
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	207.20	32.59	225.43	1,079.91
Nutrient balance	81.10	-2.49	49.05	235.44
Applied to removed ratio	1.39	0.92	1.22	1.22

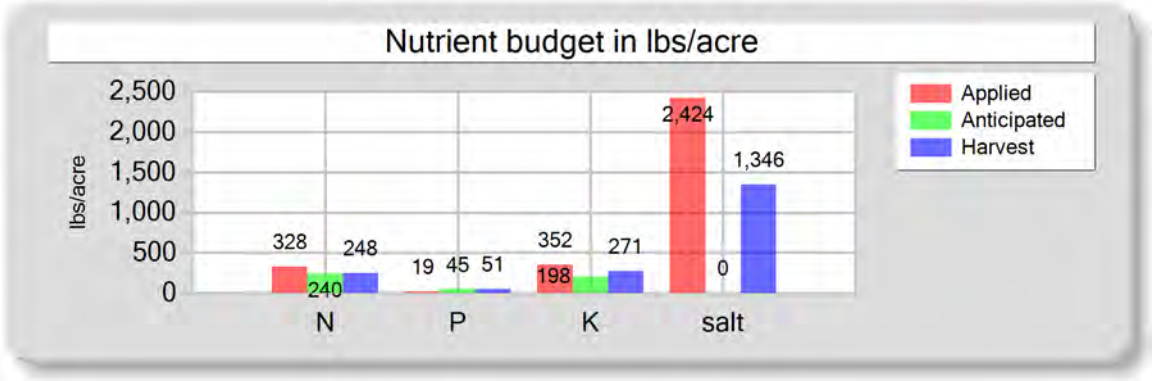
Fresh water applied	
23,460,000.00	*gallons*
863.95	*acre-inches*
11.52	*inches/acre*

Process wastewater applied	
2,741,551.00	*gallons*
100.96	*acre-inches*
1.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

B7 - 05/31/2023: Corn, silage

Field name: B7 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	321.00	18.57	351.74	2,133.71
Fresh water	0.00	0.00	0.00	290.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	328.00	18.57	351.74	2,423.74
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	247.57	50.89	270.90	1,346.11
Nutrient balance	80.43	-32.33	80.84	1,077.63
Applied to removed ratio	1.32	0.36	1.30	1.80

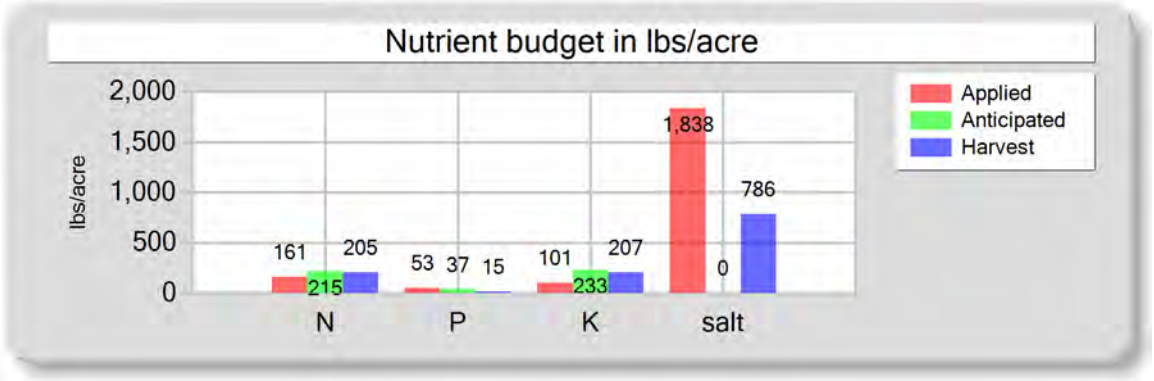
Fresh water applied	
68,595,000.00	*gallons*
2,526.12	*acre-inches*
33.68	*inches/acre*

Process wastewater applied	
2,725,224.00	*gallons*
100.36	*acre-inches*
1.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

B8 - 02/12/2019: Almond, in shell

Field name: B8 Crop: Almond, in shell Plant date: 02/12/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	76.80	0.00	0.00	0.00
Dry manure	70.52	52.80	101.43	1,570.11
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	267.58
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	161.32	52.80	101.43	1,837.68
Anticipated crop nutrient removal	215.00	36.50	232.50	0.00
Actual crop nutrient removal	205.34	14.58	206.95	785.55
Nutrient balance	-44.03	38.22	-105.52	1,052.13
Applied to removed ratio	0.79	3.62	0.49	2.34

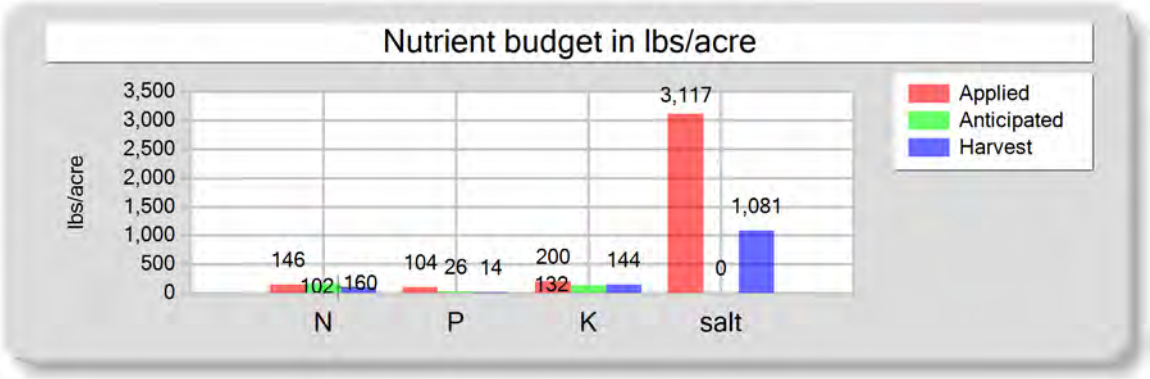
Fresh water applied	
39,189,689.00	*gallons*
1,443.22	*acre-inches*
37.98	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

C1 - 11/18/2022: Oats, hay

Field name: C1 Crop: Oats, hay Plant date: 11/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	139.25	104.27	200.30	3,100.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	16.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.25	104.27	200.30	3,116.68
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	102.39	14.12	143.73	1,080.82
Nutrient balance	43.86	90.15	56.57	2,035.86
Applied to removed ratio	1.43	7.39	1.39	2.88

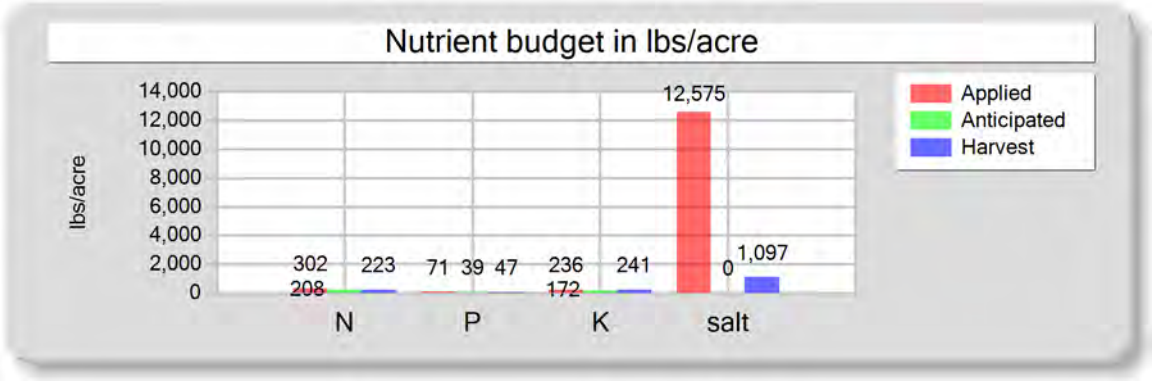
Fresh water applied	
27,540,000.00	*gallons*
1,014.20	*acre-inches*
11.40	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

C1 - 06/21/2023: Corn, silage

Field name: C1 Crop: Corn, silage Plant date: 06/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	231.18	70.74	236.40	12,445.53
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	129.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.18	70.74	236.40	12,574.83
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	223.21	47.10	240.88	1,097.11
Nutrient balance	78.97	23.64	-4.48	11,477.72
Applied to removed ratio	1.35	1.50	0.98	11.46

Fresh water applied	
106,080,000.00	*gallons*
3,906.57	*acre-inches*
43.89	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/15/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,300.00	15,200.00	29,200.00							45.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 15.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	4,070.00	13,600.00							71.60
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 05/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,100.00	3,790.00	13,000.00							27.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 19.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,500.00	10,200.00	48,000.00							45.50
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	4,590.00	8,500.00							36.40
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	470.00	286.00	0.00	0.00	113.00	703.00								5,520.00	2,720
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 10/25/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,920.00	547.00	0.00	0.00	131.00	640.00								4,470.00	4,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	988.00	378.00	0.00	0.00	92.30	940.00								7,080.00	4,980
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,540.00	546.00	0.00	0.00	82.40	1,160.00	224.00	157.00	292.00	4,700.00	0.00	482.00	416.00	7,870.00	5,100
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	501.00	333.00	0.00	0.00	36.70	1,160.00								6,670.00	9,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 10/13/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	818.00	126.00	0.00	0.00	66.40	700.00								4,690.00	1,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,110.00	543.00	0.00	0.00	120.00	1,180.00								9,670.00	7,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

MID Lateral 4

MID Lateral 4

Sample description: MID Lateral 4

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

MID Lateral 4

MID Lateral 4

Sample description: MID Lateral 4

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								37.60	38
DL	1.00	0.50	0.50								10.00	10

TID Turlock Main

TID Turlock Main

Sample description: TID Turlock Main

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

TID Turlock Main

Sample description: TID Turlock Main

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.30	13
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

A2 - 10/27/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,650.00	830.00	4,940.00		11.70
DL	100.00	100.00	100.00		0.01

A2 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,260.00	830.00	4,000.00		5.10
DL	100.00	100.00	100.00		0.01

B1 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	850.00	6,750.00		11.70
DL	100.00	100.00	100.00		0.01

B1 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,930.00	800.00	6,130.00		5.00
DL	100.00	100.00	100.00		0.01

B2 - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	8,500.00	67,500.00		11.70
DL	100.00	100.00	100.00		0.01

B2 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,930.00	800.00	6,130.00		5.00
DL	100.00	100.00	100.00		0.01

B3 - 11/04/2002: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	850.00	6,750.00		11.70
DL	100.00	100.00	100.00		0.01

B3 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,930.00	800.00	6,130.00		5.00
DL	100.00	100.00	100.00		0.01

B4 - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	850.00	6,750.00		11.70
DL	100.00	100.00	100.00		0.01

B4 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,930.00	800.00	6,130.00		5.00
DL	100.00	100.00	100.00		0.01

B5 - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	850.00	6,750.00		11.70
DL	100.00	100.00	100.00		0.01

B5 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,930.00	800.00	6,130.00		5.00
DL	100.00	100.00	100.00		0.01

B6 - 11/04/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,640.00	850.00	6,750.00		11.70
DL	100.00	100.00	100.00		0.01

B6 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,930.00	800.00	6,130.00		5.00
DL	100.00	100.00	100.00		0.01

B7 - 10/26/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,340.00	840.00	5,810.00		9.80
DL	100.00	100.00	100.00		0.01

B7 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,670.00	960.00	5,110.00		6.90
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,330.00	780.00	6,240.00		22.80
DL	100.00	100.00	100.00		0.01

B8 - 02/12/2019: Almond, in shell

Almonds

Sample and source description: Almonds
Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 6.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,500.00	1,810.00	25,700.00		10.40
DL	100.00	100.00	100.00		0.01

C1 - 11/18/2022: Oats, hay

C1 - 11/18/2022: Oats, hay

Oat Hay

Sample and source description: Oat Hay

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,050.00	1,110.00	11,300.00		9.40
DL	100.00	100.00	100.00		0.01

C1 - 06/21/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,170.00	880.00	4,500.00		5.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

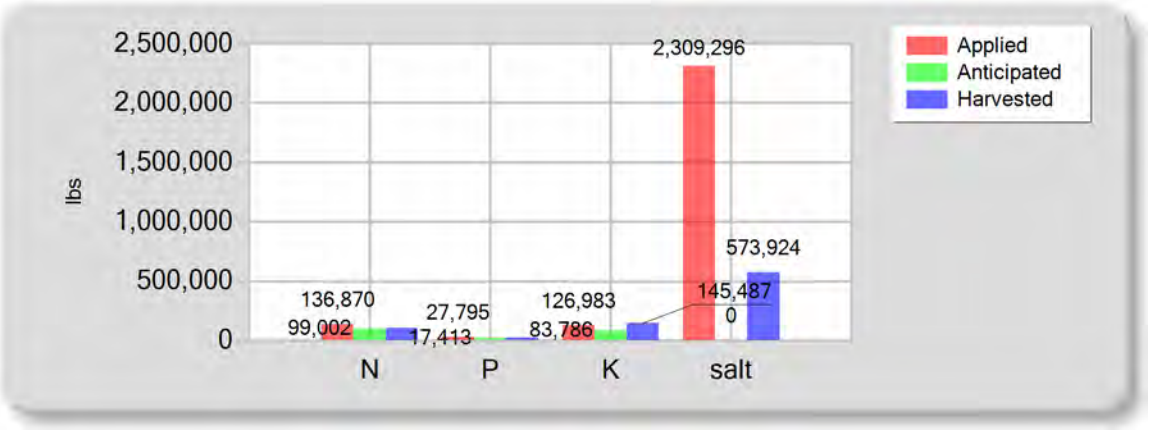
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

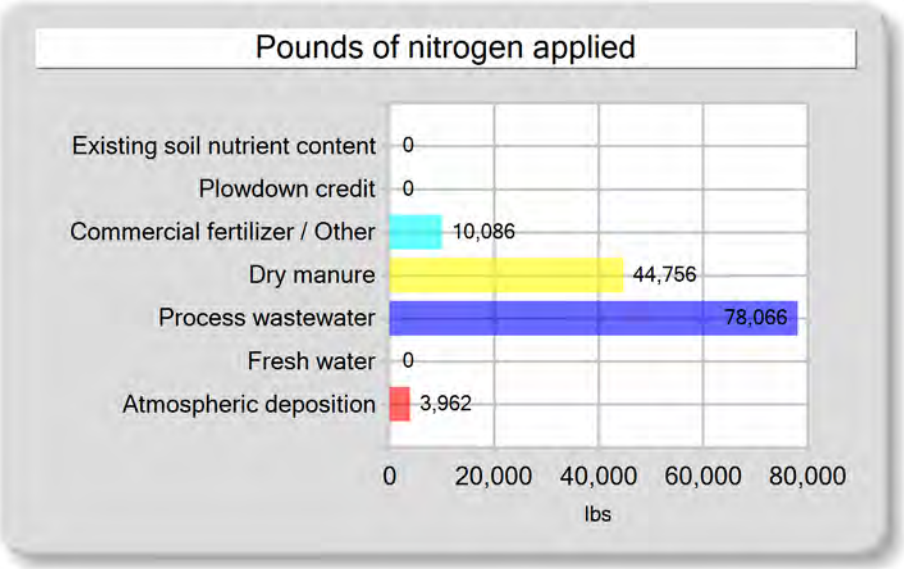
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

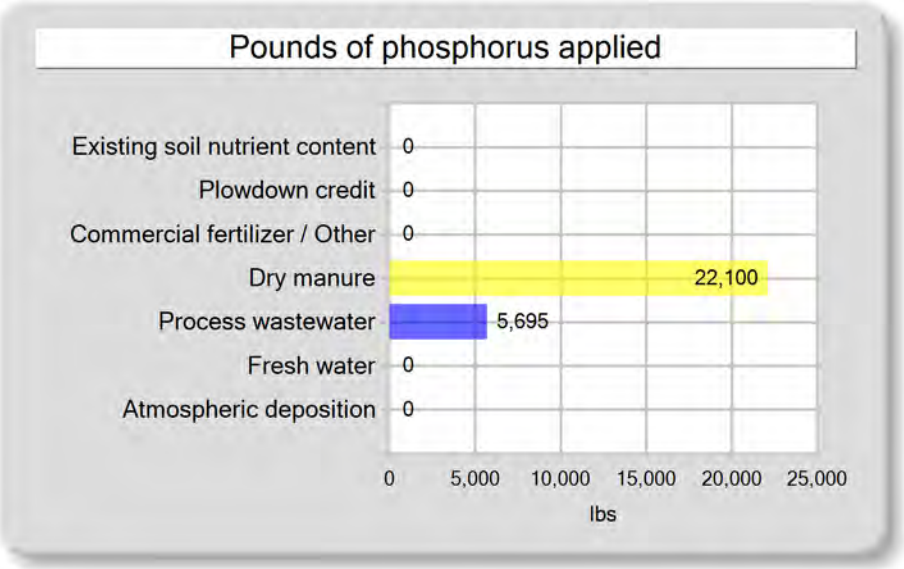
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10,086.40	0.00	0.00	0.00
Dry manure	44,755.90	22,099.66	53,658.80	1,810,228.00
Process wastewater	78,065.72	5,695.32	73,323.96	418,541.23
Fresh water	0.00	0.00	0.00	80,527.27
Atmospheric deposition	3,962.00	0.00	0.00	0.00
Total nutrients applied	136,870.02	27,794.98	126,982.76	2,309,296.50
Anticipated crop nutrient removal	99,002.00	17,412.50	83,786.40	0.00
Actual crop nutrient removal	104,560.01	21,237.17	145,487.20	573,923.59
Nutrient balance	32,310.01	6,557.81	-18,504.44	1,735,372.91
Applied to removed ratio	1.31	1.31	0.87	4.02

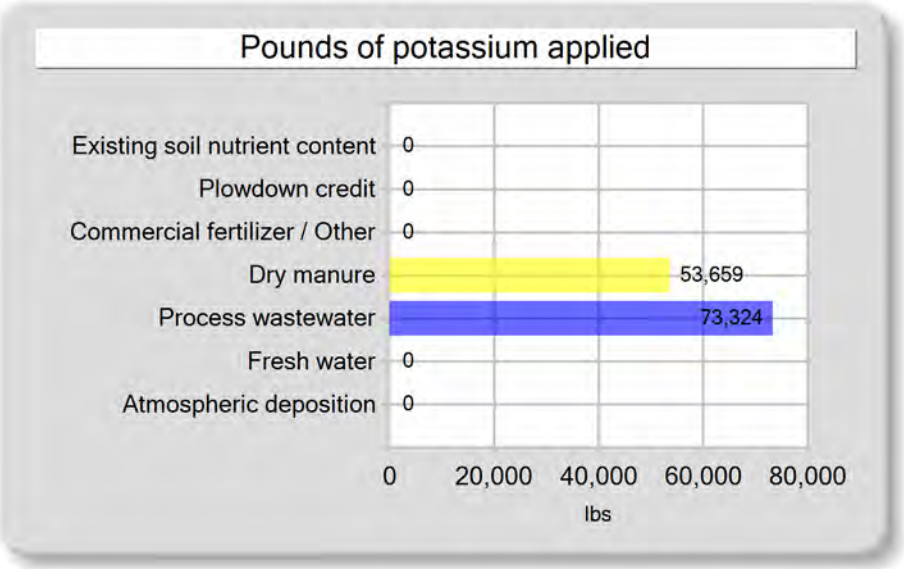
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

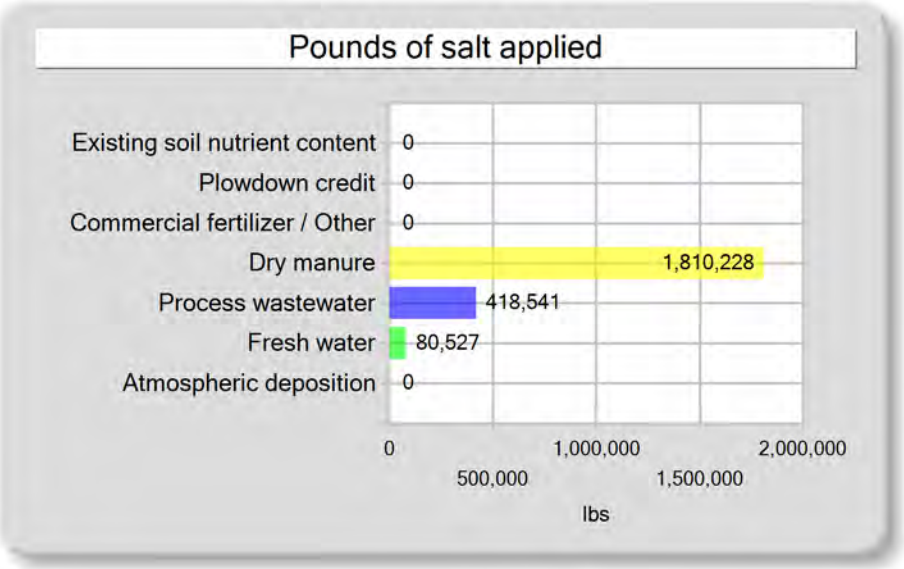


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Frank Gwerder (Jun 21, 2024 21:22 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Gwerder	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 21, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: FRANK GWERDER DAIRY INC.

Facility Address: 1007 S. HAIRT RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person Name and Phone Number: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: AVILA SPREADING SERVICE

Address of Hauling Company /Person: P.O. Box 580742 (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person: LONNY AVILA (Name) 209-595-7955 (Phone Number)

Destination Information:

Composting Facility / (Broker) / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

LONNY AVILA (Name) P.O. Box 580742 (Number and Street) MODESTO (City) 95358 (Zip Code) 209-595-7955 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

7243 PARADISE RD. (Number and Street) MODESTO (City) 95358 (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 8/30/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1640 (Tons) or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 19.6% MOISTURE

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

164 LOADS @ 10 TONS PER LOAD.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

Hauler's Signature: _____ Date: _____

AVILA

PROFESSIONAL · RELIABLE · COURTEOUS

SPREADING SERVICE

P.O. Box 580742 Modesto, California 95358

Frank Gwerder Dairy
1007 S Hart Rd
Modesto, Ca. 95358
Panchomoo@aol.com

Date: 8-30-23
Terms: 10 Days
Fed ID: 77-0480211

Loads	Tons	Yards	Description	Cost
164	1640	3280	Manure taken by Avila Spreading Service July/ August	

*Manure from Dairy 1, Hart Rd

Total:

Past due after 10 days from date. If account unpaid within time permitted, customer agrees to pay interest at 18% per annum and attorney's fees if account is referred to an attorney or collection agency for collection.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

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3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: FRANK GWERDER DAIRY INC.

Facility Address: 1007 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person Name and Phone Number: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: FRANK GWERDER

Address of Hauling Company /Person: 825 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

IRV HILARIO (Name) 506 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code) 209-527-3807 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 012-050-005 (Assessor's Parcel Number)

Dates Hauled: 4/9/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 812,596 Gallons

Method used to determine volume of process wastewater: _____

FLOWMETER

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

Hauler's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

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2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: FRANK GWERDER DAIRY INC.

Facility Address: 1007 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person Name and Phone Number: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: FRANK GWERDER

Address of Hauling Company /Person: 825 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

IRV HILARIO (Name) 506 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code) 209-527-3807 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 012-050-005 (Assessor's Parcel Number)

Dates Hauled: 5/25/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 956,462 Gallons

Method used to determine volume of process wastewater: _____

FLOWMETER

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Frank Gweider (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

Hauler's Signature: Frank Gweider (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

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4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: FRANK GWERDER DAIRY INC.

Facility Address: 1007 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person Name and Phone Number: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: FRANK GWERDER

Address of Hauling Company /Person: 825 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

IRV HILARIO (Name) 506 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code) 209-527-3807 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 012-050-005 (Assessor's Parcel Number)

Dates Hauled: 7/22/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 934,587 Gallons

Method used to determine volume of process wastewater: _____

FLOWMETER

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

Hauler's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

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Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: FRANK GWERDER DAIRY INC.

Facility Address: 1007 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person Name and Phone Number: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: FRANK GWERDER

Address of Hauling Company /Person: 825 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

IRV HILARIO (Name) 506 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code) 209-527-3807 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street / City / Zip Code / Assessor's Parcel Number: 012-050-005

Dates Hauled: 8/4/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 917,626 Gallons

Method used to determine volume of process wastewater: _____

FLOWMETER

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

Hauler's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

ATTACHMENT D

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For
Existing Milk Cow Dairies

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4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: FRANK GWERDER

Name of Dairy Facility: FRANK GWERDER DAIRY INC.

Facility Address: 1007 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person Name and Phone Number: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: FRANK GWERDER

Address of Hauling Company /Person: 825 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code)

Contact Person: FRANK GWERDER (Name) 209-579-5723 (Phone Number)

Destination Information:

Composting Facility / Broker / (Farmer) / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

IRV HILARIO (Name) 506 S. HART RD. (Number and Street) MODESTO (City) 95358 (Zip Code) 209-527-3807 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street _____ City _____ Zip Code _____ Assessor's Parcel Number 012-052-008

Dates Hauled: 10/23/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 712,500 Gallons

Method used to determine volume of process wastewater: _____

75 LOADS OF SLURRY @ 9500 GALLONS PER LOAD.

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

✓ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

Hauler's Signature: Frank Gwerder (Jun 21, 2024 21:22 PDT) Date: Jun 21, 2024

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Frank Gwerder Dairy Inc.
1007 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - #1 Main Dairy	T3082014A	Electrical Conductivity (EC)	EPA 120.1 and	1.14	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	24.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	731	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Phone (209) 634-9055 - Fax (209) 634-9057

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Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Frank Gwerder Dairy Inc.
1007 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - #2 Employee	T3082014B	Electrical Conductivity (EC)	EPA 120.1 and	1.48	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	41.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	949	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Frank Gwerder Dairy Inc.
1007 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - #3 Heifers	T3082014C	Electrical Conductivity (EC)	EPA 120.1 and	0.255	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.70	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	163	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101			
Electrical Conductivity (EC)	102			3.27
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

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DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: FRANK GWERDER
Mailing address:
(not facility address) MODESTO CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	3/23/23	12:40	DW #1 MAIN DAIRY	DOMESTIC	19.2		X				
2		12:45	DW #2 EMPLOYEE		19.4		X				
3		12:50	DW #3 HEIFERS		19.4		X				
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-23-23 3:30			T3082014

Required to be submitted to any agency? (YES / NO)

2023 Frank Gwerder Annual Report

Final Audit Report 2024-06-22

Created:	2024-06-21
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAwR-ZwIlck6fzDAYMwCGyk2nkeSCxfuwC

"2023 Frank Gwerder Annual Report" History

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